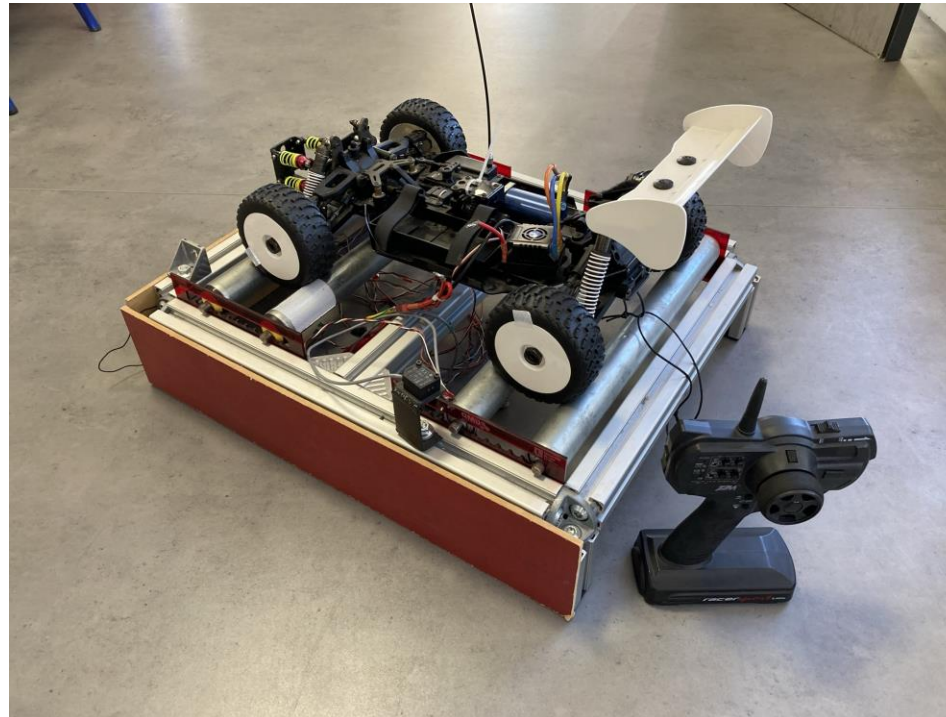


Revue de Projet

Instrumentation d'un banc d'essai pour voitures Radio commandées (RC)



*FVRC
Mini RC de Marigny*

*Année 2025-2026
Professeur responsable :
Monsieur BEAU*

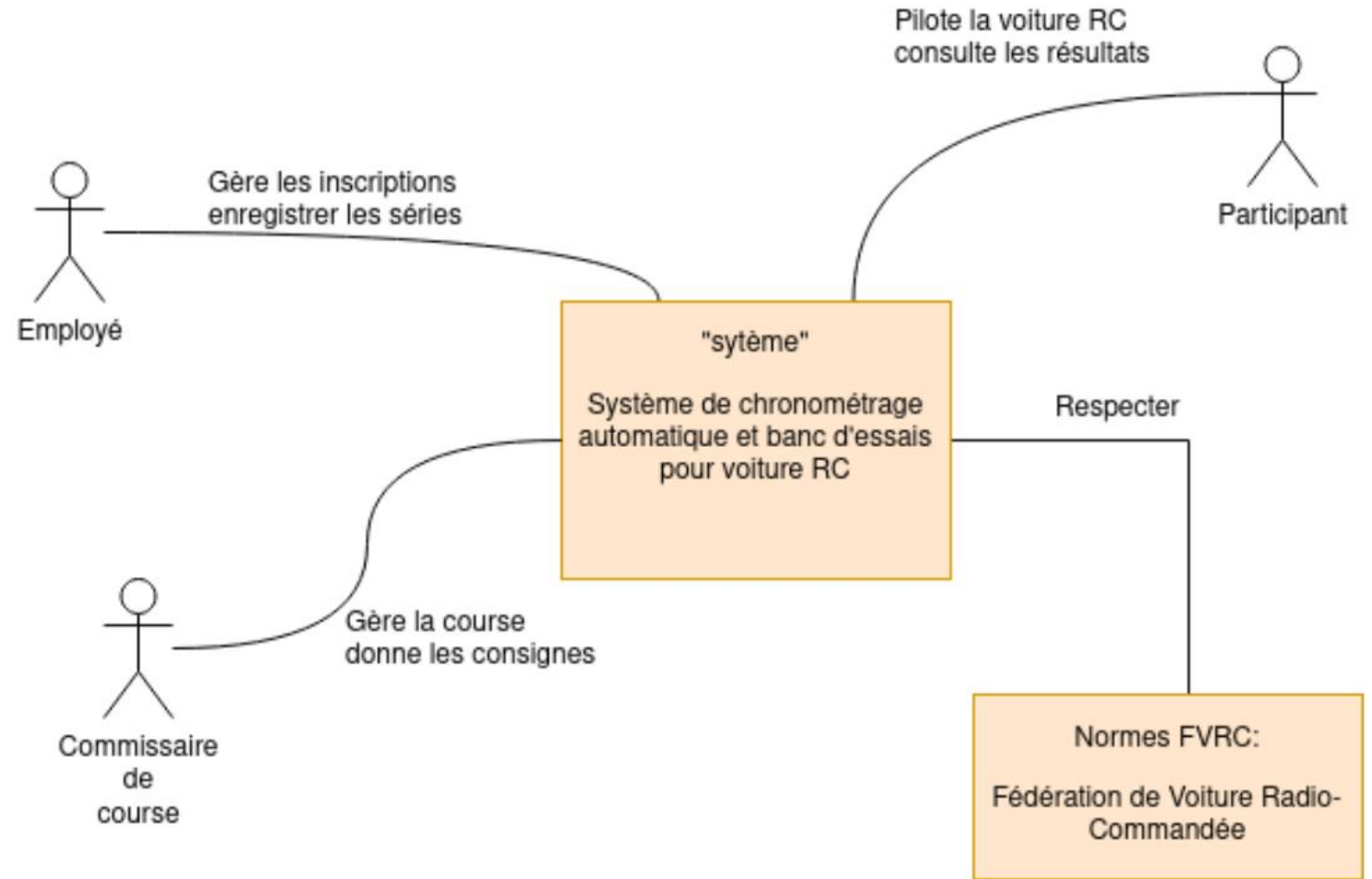
*Etudiant chargés du projet :
AIT AMIR Youssouf
DUCHATEAU Edouard*



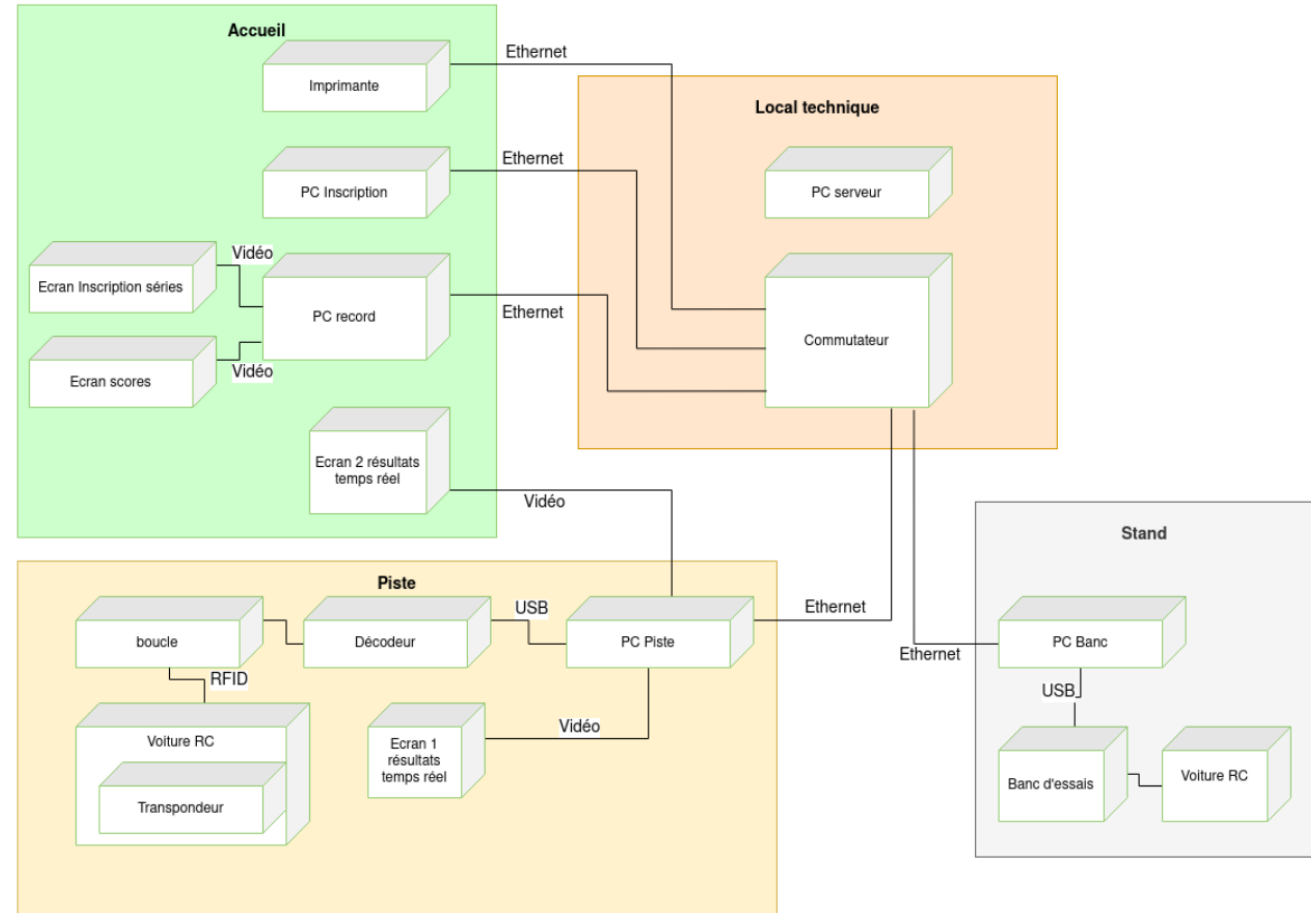
Sommaire

- Contexte Général / Applicatif
- Objectifs du banc d'essai
- Identification du besoin
- Matériel du projet
- Module NFC
- Serveur Raspberry Pi
- Montage final du projet
- Code
- Avant la revue finale
- Etape proactive du projet
- Conclusion

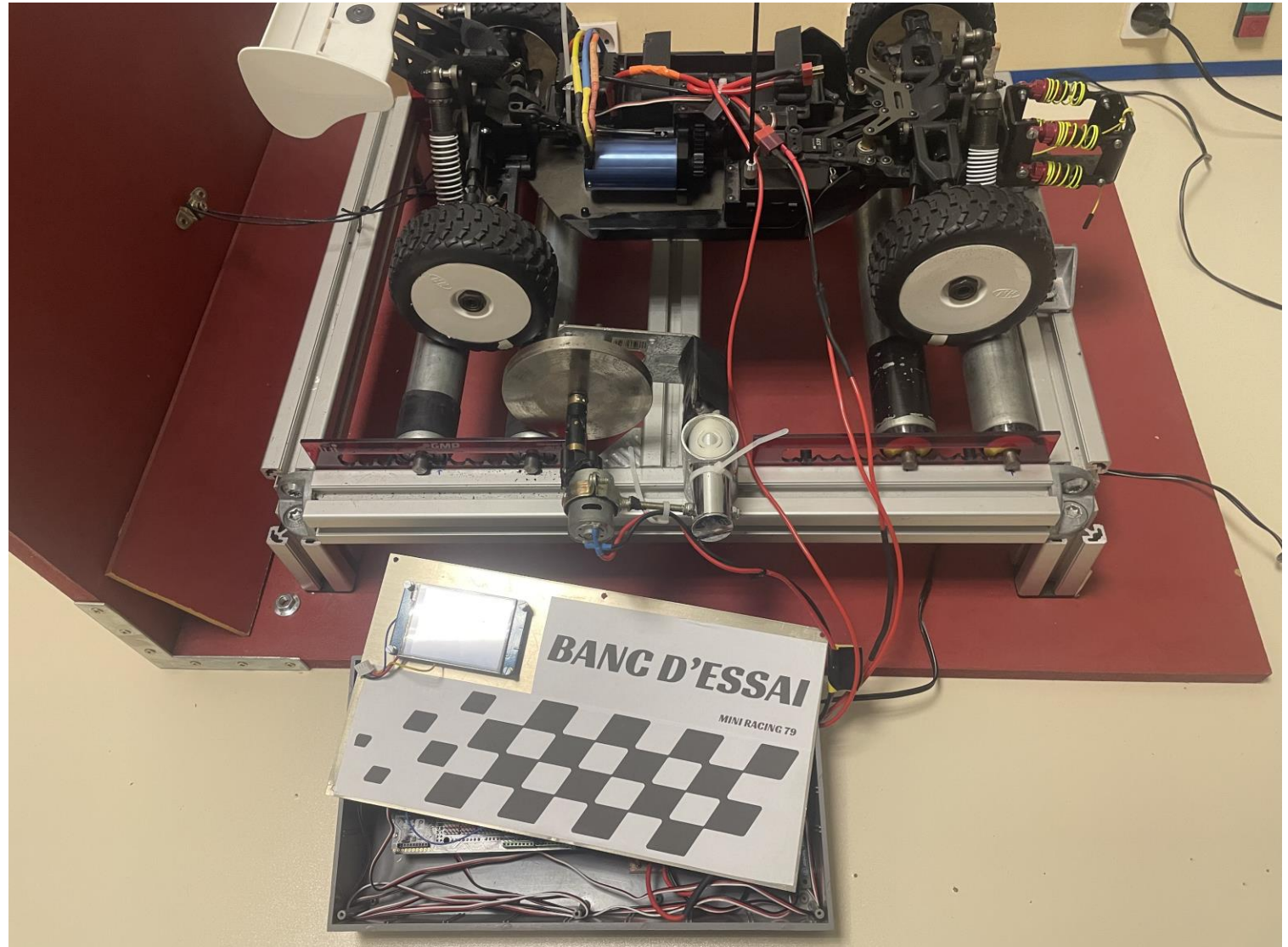
Contexte Général du projet



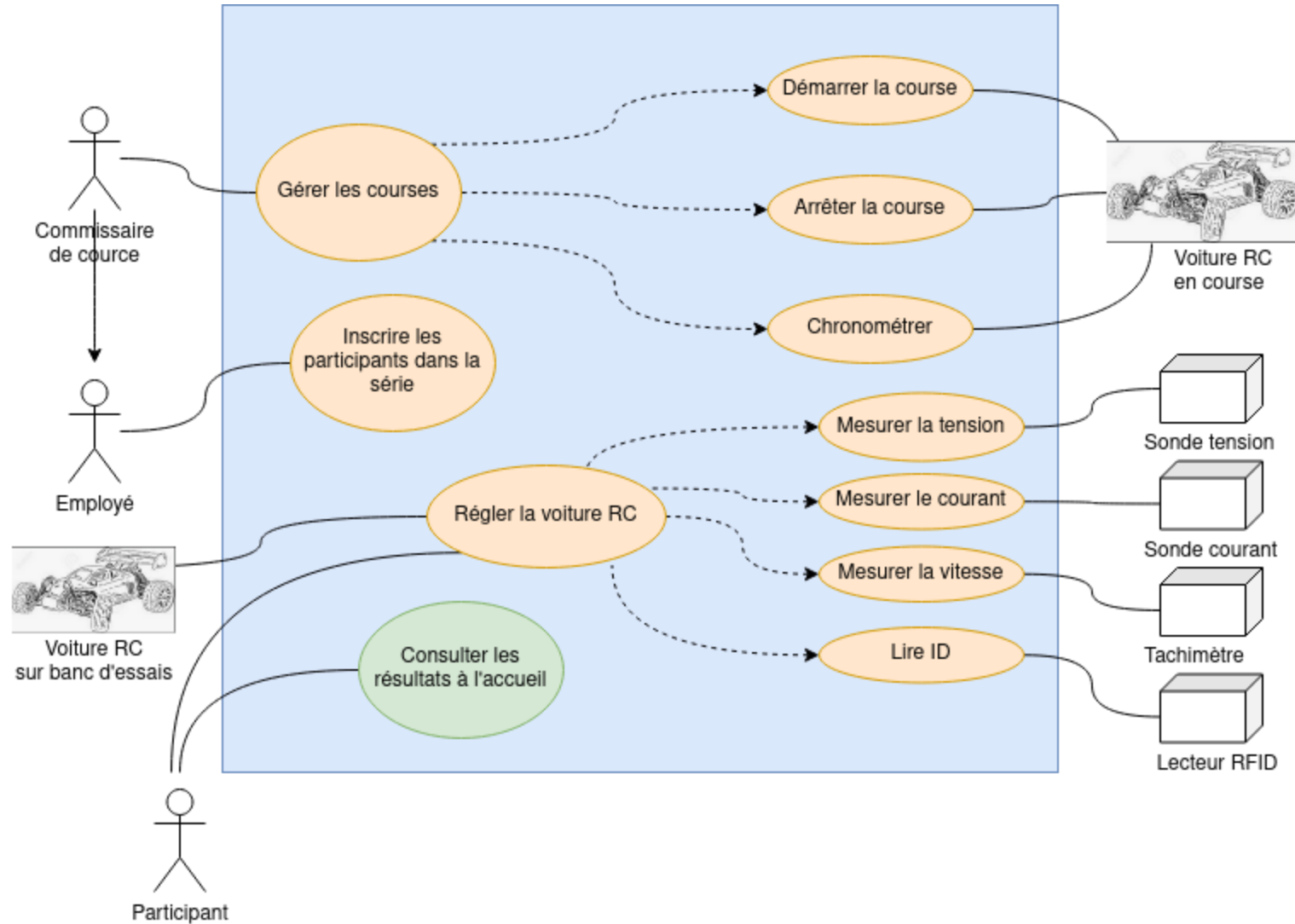
Contexte Applicatif : Club et environnement



Objectifs du Banc d'essai RC

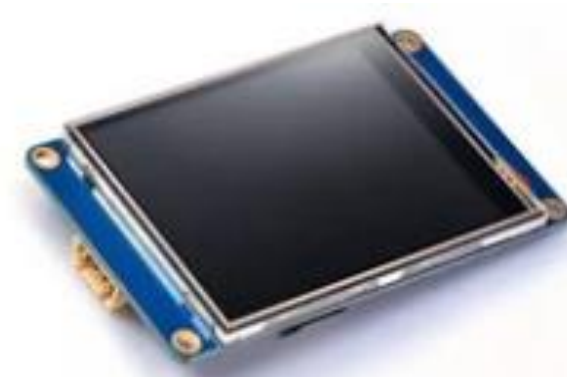
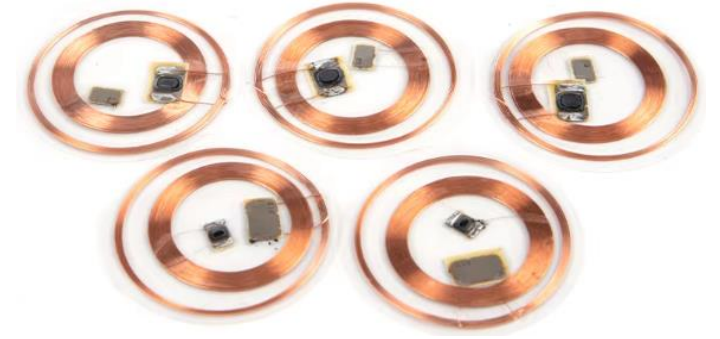
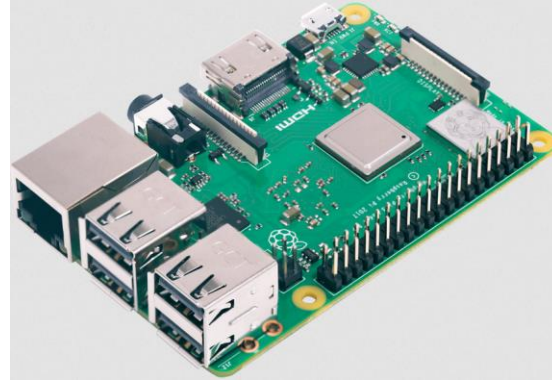


Identification du besoin





Matériel du projet

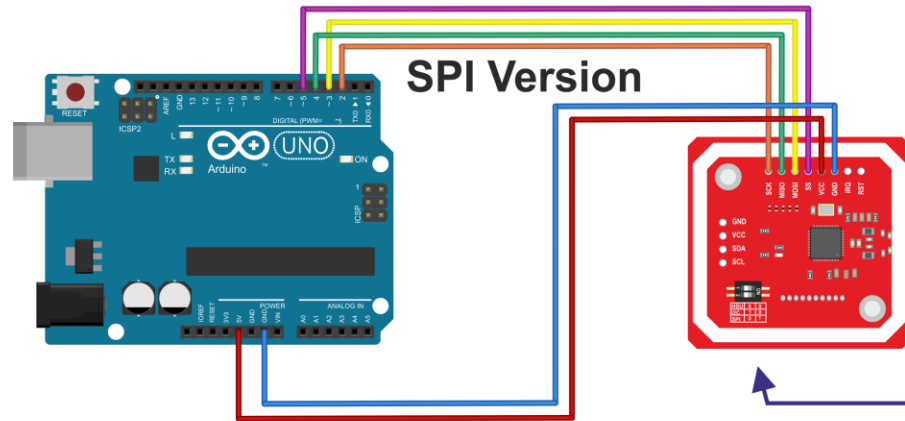


Module NFC



PN532 NFC RFID Modul V3

RFID Seite

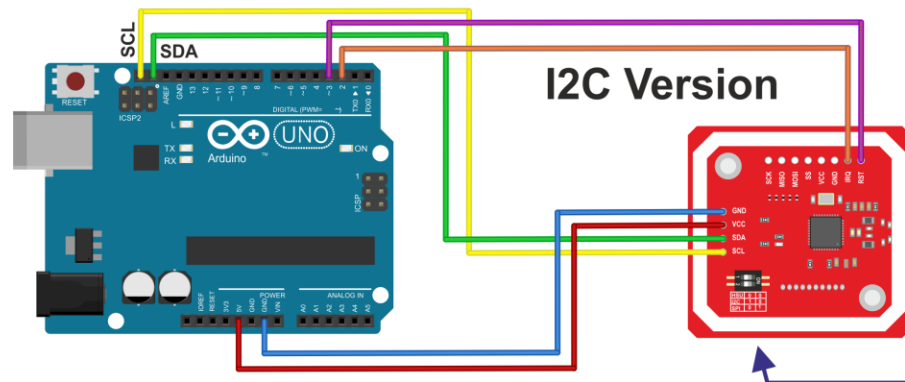


SPI Version

SPI Version



HSU	0	0
I2C	1	0
SPI	0	1



I2C Version

I2C Version

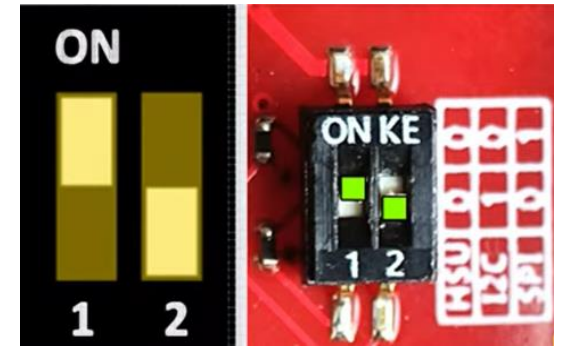


HSU	0	0
I2C	1	0
SPI	0	1

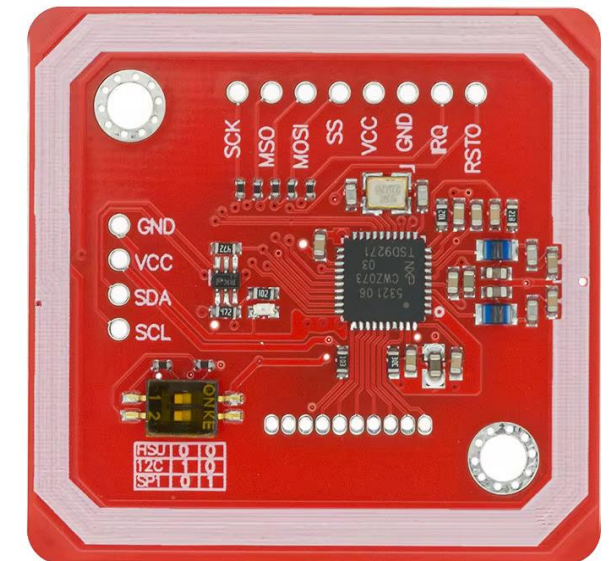
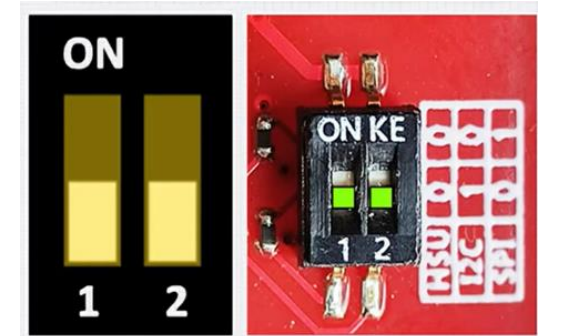
Possible protocols :

- I2C
- SPI
- UART (HSU)

I2C



UART





Paramètres généraux

Modifier le mot de passe

Interclassement pour la connexion au serveur : utf8mb4_unicode_ci

Plus de paramètres

Paramètres d'affichage

Langue (Language) Français - French

Thème pmahomme Tout afficher

utilisateurs	performances
id	uid_rc
nom	distance
prenom	vitesse
	roue_tr/s
	state

Colonne	Type	Fonction	Null	Valeur
id	int(11) unsigned			
nom	varchar(50)			
prenom	varchar(50)			

Exécuter

```
sudo apt install apache2
sudo apt install php libapache2-mod-php
```

```
cd /var/www/html
sudo nano test.php
<?php
phpinfo();
?>
```

Serveur de base de données

- Serveur : Localhost via UNIX socket
- Type de serveur : MariaDB
- Connexion au serveur : SSL n'est pas utilisé
- Version du serveur : 11.8.6-MariaDB-0+deb13u1 from Debian - -- Please help get to 10k stars at <https://github.com/MariaDB/Server>
- Version du protocole : 10
- Utilisateur : root@localhost
- Jeu de caractères du serveur : UTF-8 Unicode (utf8mb4)

Serveur Web

- Apache/2.4.66 (Debian)
- Version du client de base de données : libmysql - mysqlnd 8.4.16
- Extension PHP : mysqli curl mbstring sodium
- Version de PHP : 8.4.16

phpMyAdmin

- Version : 5.2.2deb1+deb13u1

Détails de l'adresse IP

IP:

100.68.68.69

Ping:

92 ms

Nom d'hôte:

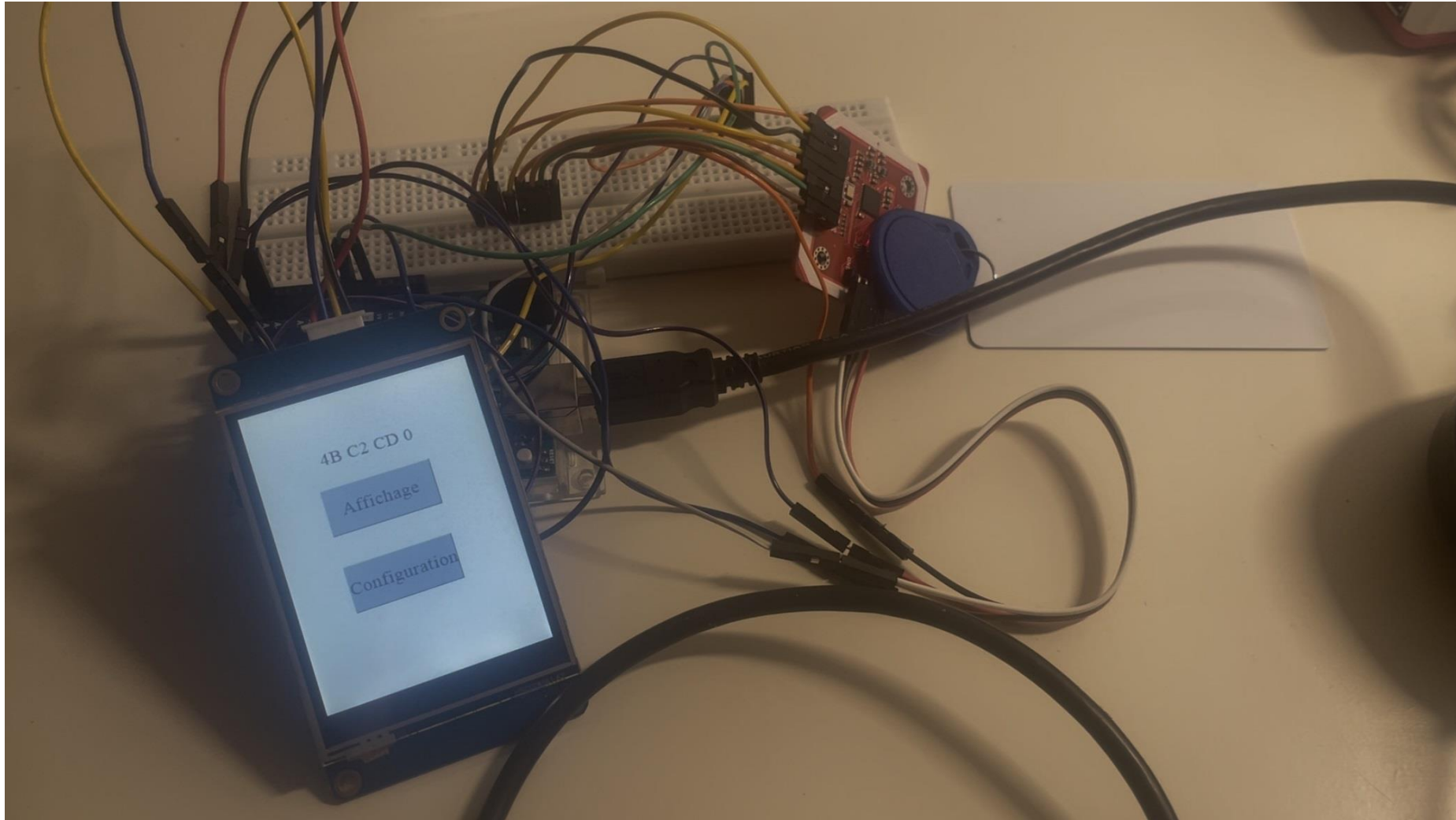
projetc.tailbedb56.ts.net.

Ports:

80

Tailscale Pi 3+

Montage final : module RFID I2C & Nextion UART



Affichage console et code

```
Output  Serial Monitor X
Message (Enter to send message to 'Arduino UNO' on 'COM5')

System init...
Found chip PN532
Firmware ver. 1.6
NFC ready
Scan card...
Tag is not NDEF formatted.
Card detected
UID:
4B C2 CD 01
```

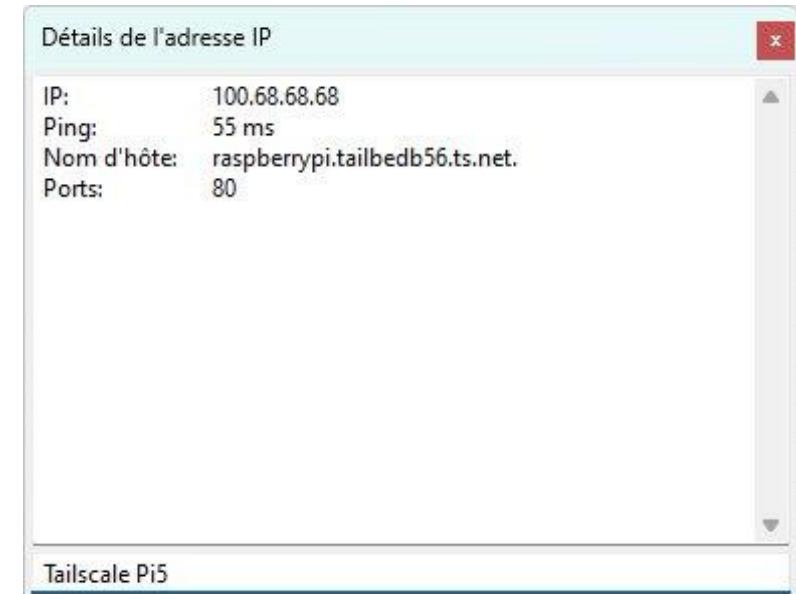
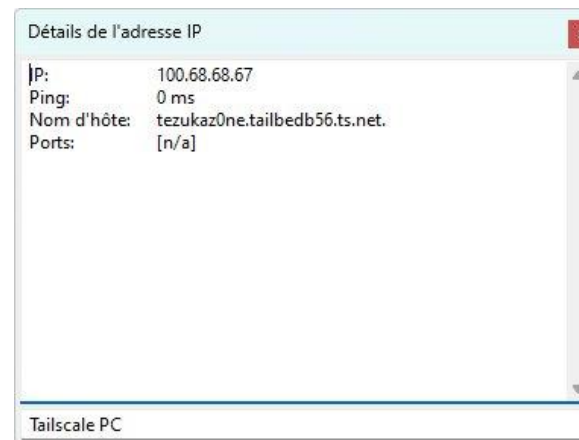
```
NFC_I2C-OLED_Mode.ino
55 // ----- Setup -----
56 void setup() {
57     Serial.begin(115200);
58     oledSerial.begin(9600);
59
60     delay(500);
61
62     // Reset écran (supprime anciens boutons affichés)
63     sendToNextion("rest");
64     delay(1000);
65
66     clearScreen();
67
68     addLine("System init...");
69
70     // NFC init
71     nfc.begin();
72     addLine("NFC ready");
73     addLine("Scan card...");
74 }
75
76 // ----- Loop -----
77 void loop() {
78
79     if (nfc.tagPresent()) {
80         NfcTag tag = nfc.read();
81
82         String uid = tag.getUidString();
```

Ajout d'un VPN "mesh" au projet



Token →

```
admin@ProjetRC: ~  
Building dependency tree... Done  
Reading state information... Done  
tailscale is already the newest version (1.96.4).  
tailscale-archive-keyring is already the newest version (1.35.181).  
The following packages were automatically installed and are no longer required:  
  alacarte gir1.2-gmenu-3.0 gnome-menus gtk-nop libgnome-menu-3-0 retry  
Use 'sudo apt autoremove' to remove them.  
0 upgraded, 0 newly installed, 0 to remove and 45 not upgraded.  
+ [ false = true ]  
+ set +x  
Installation complete! Log in to start using Tailscale by running:  
  
sudo tailscale up  
admin@ProjetRC:~ $ sudo tailscale up  
  
To authenticate, visit:  
  
    https://login.tailscale.com/a/865beec01fef5  
  
Success.  
admin@ProjetRC:~ $ tailscale ip  
100.64.216.101  
fd7a:115c:a1e0:d73a:d865  
admin@ProjetRC:~ $
```



Rendu Web

y.tzivanis@gmail.comFree

?

Y

Machines

Apps

Services

Users

Access controls

Logs

DNS

Settings

Resource hub

Machines

Manage the devices connected to your tailnet. [Learn more](#)

Add device

Search by name, owner, tag, version...

Filters

[Learn more](#)

3 machines

MACHINE	ADDRESSES	VERSION	LAST SEEN	
projetc y.tzivanis@gmail.com SSH	100.68.68.69	1.96.4 Linux 6.12.75+rpt-rpi-v8	Connected	
raspberrypi y.tzivanis@gmail.com	100.68.68.68	1.96.4 Linux 6.12.47+rpt-rpi-v8	Connected	
tezukazOne y.tzivanis@gmail.com	100.68.68.67	1.96.3 Windows 11 23H2	Connected	

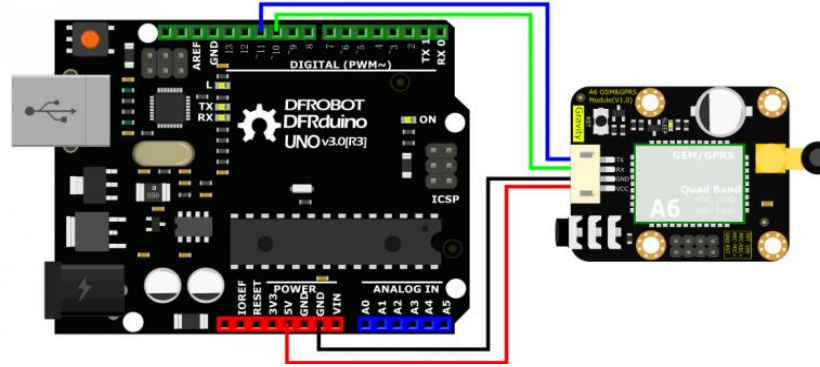


LM393
with PWM



Phase de consolidation avant la revue finale

Etape proactive : ajout d'un module complémentaire



A6 GSM Module :
Send and
receive sms

- W-freq: quad-band network, 850 / 900 / 1800 / 1900MHz
- W-Voltage: 4.8-9VDC
- W-Current: max 2A
- Communication Interface: TTL serial port / RS232 serial port
- Baud rate: 115200bps

